Hafsa Sheikh

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Education

ETH Zurich Sept. 2025 – ongoing
MSc Computer Science *Zurich, Switzerland*

- **Major: Data Management Systems, Minor: Programming Languages & Software Engineering.** Coursework: Big Data, Cloud Computing Architecture, Advanced Systems Lab, Concepts of Object-Oriented Programming

ETH Zurich Sept. 2022 – Sept. 2025
BSc Computer Science - GPA: 5.15/6 *Zurich, Switzerland*

- **Thesis (6/6), Secure and Trustworthy Systems Group:** extended XRAY, a hardware security analysis tool; an automated pipeline ran **300,000+ test cases** and uncovered previously unknown vulnerabilities.
- Coursework: Algorithms and Data Structures, Operating Systems, Compiler Design, Parallel Programming, Data Modeling and Databases, Computer Networks

Experience

ETH Zurich Feb – Jun 2024, 2025, 2026
Teaching Assistant - Digital Design and Computer Architecture *Zurich, Switzerland*

- Guided students through **10 hands-on lab sessions** (Verilog, MIPS assembly, FPGA) and reviewed their submissions.

Projects

Real or AI? | *TypeScript, Astro, Svelte, Hono, Cloudflare Workers, D1, R2* | realoraiquiz.com

- **Built and launched a web game** where players guess whether an image is a real photo or AI-generated: daily challenge, timed Blitz mode, leaderboards, accounts and comments.
- **Automated content pipeline**: a vision model describes each public-domain photo and FLUX generates a matching AI twin; both are normalised so files cannot leak the answer, and new pairs arrive nightly via GitHub Actions.
- Server-side scoring, per-IP rate limits, hashed one-time sign-in codes, strict CSP and AI comment moderation; **120+ unit and API tests plus Playwright end-to-end tests** run in CI before every automatic deploy.

KnowHouse | *Flutter, Dart, Firebase, GitHub Actions, HTML, CSS* | getknowhouse.com | Google Play

- **Built and shipped a family task app on Google Play**: task assignment, real-time collaboration, leaderboards, shared calendar, personalised task recommendations and user profiles.
- Wrote **200+ unit, provider and widget tests**; set up **CI/CD with GitHub Actions** that versions, signs and uploads every release to Google Play automatically.

Shax: Somali Strategy Board Game | *Flutter, Dart, Game AI, i18n, HTML, CSS* | shax.app

- Built a **minimax game AI** with three difficulty levels and the full traditional rule set; **one codebase for Android, iOS and web** in three languages, with automated signed releases.

Compiler for an Object-Oriented Language | *OCaml, LLVM IR, x86-64*

- Built a compiler for Oat from scratch: **lexing, parsing and type checking**, and a backend from **LLVM IR to x86-64**.
- Implemented an x86-64 simulator and assembler, **dataflow optimisations and register allocation**.

FastGATr: High-Performance Transformer Kernels | *C++, ARM NEON SIMD, Python*

- Optimised a geometric algebra transformer in C++ on Apple M1 (generated sparse kernels, **~140x fewer ops**, cache-aware repacking, NEON intrinsics): **1500x over naive C++** and **1.7-4.5x over the optimised PyTorch reference**.

Skills

Languages: C++, Java, Python, C, JavaScript, TypeScript, SQL, Dart, OCaml, HTML, CSS, Verilog/SystemVerilog
Tools: Flutter, React, Svelte, Firebase, Cloudflare Workers, PostgreSQL, Kubernetes, Docker, GitHub Actions, LLVM
Spoken: German (native), English (fluent), French (basic)